

# Saad Driouech

Fürth, Germany (open to relocation)  
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## Profile Summary

Machine Learning Engineer with a research background in generative AI and applied ML. M.Sc. thesis introduced the first spatial control framework for scale-wise autoregressive T2I generation. Experience spans diffusion models, autoregressive generation, RAG pipelines, and AI automation. Published 5 peer-reviewed papers. FAU grade 1.2 (Top 1%).

## Experience

**Fraunhofer IIS** Nuremberg, Germany  
*Generative AI Research Engineer (Working Student)* 03/2025 – Present

- Training diffusion-based generative models for GNSS signal reconstruction, iterating on architectures and loss functions to stabilise training and prevent mode collapse.
- Designed evaluation frameworks to measure robustness under noise and distribution shift; experimented with spectrogram and complex IQ data representations; tracked experiments with TensorBoard.

**August-Wilhelm Scheer Institut** Saarbrücken, Germany  
*Applied Machine Learning Engineer (Working Student)* 12/2023 – 02/2025

- Built end-to-end ML pipelines for garment return prediction, achieving **86% balanced accuracy** on cold-start products; refactored preprocessing for a **5x speedup**.

**Hightech Payment Systems** Casablanca, Morocco  
*Development Engineer* 06/2023 – 08/2023

- Enhanced PowerCARD payment switching platform for VISA/Mastercard compliance; worked with SQL, Docker, CI/CD pipelines, and Linux in a professional engineering environment.

**Al Akhawayn University** Ifrane, Morocco  
*Applied NLP Research Engineer (Part-time)* 09/2022 – 05/2023

- Pre-trained transformer language models (52M–80M params) for low-resource Moroccan Darija; fine-tuned on text summarization (ROUGE-1 **19.25**, state-of-the-art), topic classification (**F1 0.84 / Acc 0.86**), and offensive language detection (**90% accuracy / 85% F1**).

## Education

**Friedrich-Alexander-Universität Erlangen-Nürnberg** Erlangen, Germany  
*M.Sc. Artificial Intelligence* 10/2023 – 05/2026

Grade: 1.2 (Top 1%). Thesis: *Finetuning Visual Autoregressive Models for Controllable Image Generation*. First spatial control framework for next-scale autoregressive T2I; ControlNet-style architecture over a 2.5B-parameter backbone, 6 control modalities, unified checkpoint.

**Al Akhawayn University** Ifrane, Morocco  
*B.Sc. Computer Science* 09/2018 – 07/2022

GPA: 3.83 (Summa Cum Laude). Thesis: Darija TTS using FastSpeech 2 + HiFi-GAN (MOS 3.9), published at ICDTA 2025.

## Personal Projects

**ApplAI** 2026

- Automated job application pipeline: n8n cron orchestration, FastAPI backend, Gemini 2.5 Flash scoring, Claude Sonnet for LaTeX CV/cover letter generation, Discord review gates, Notion dashboard, Docker Compose; security-first (JD sanitization, Ed25519 verification, parameterized SQL).

**PaperLens** 2026

- RAG search system: FastAPI, Qdrant, sentence-transformers, reranking, Groq (Llama-3.3-70b); custom retrieval quality evaluation metrics.

## Technical Skills

**ML & DL:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Diffusion Models, Transformers

**NLP & Speech:** LLMs, TTS (FastSpeech 2, HiFi-GAN), Fine-tuning (LoRA/PEFT), Hugging Face, spaCy

**LLM Systems:** LangChain, LangGraph, LlamaIndex, RAG, Qdrant, FAISS, OpenAI/Anthropic/Groq APIs

**Infra:** Python, FastAPI, SQL, Docker, Airflow, Git, CI/CD, Linux

**Languages:** English (Fluent), French (Advanced), Arabic (Native), German (Learning)